UNIVERSITY OF BOTSWANA
END OF SEMESTER *SUPPLEMENTARY* FINAL EXAMINATIONS
FRONT PAGE

COURSE NO: **CSI247** DURATION: **2HRS** DATE: **JANUARY 2018**

TITLE OF PAPER: **DATA STRUCTURES**

SUBJECT: **COMPUTER SCIENCE**

TITLE OF EXAMINATION: **BSc II** MORNING/AFTERNOON

INTRUCTIONS:

ANSWER **ALL** QUESTIONS

MARKS ALLOCATED TO QUESTIONS ARE INDICATED NEXT TO THE QUESTION NUMBER.

TOTAL MARKS: **50**

ACADEMIC DISHONESTY harms the good reputation of our University, affects your personal integrity, is a barrier to the acquisition of the knowledge, skills and attitudes you would have acquired at the end of your study, decreases your value in the eyes of employers, has a heavy penalty including expulsion from UB

DO NOT CHEAT OR CONDONE CHEATING BY OTHERS

DO NOT OPEN THIS PAPER UNTIL YOU HAVE BEEN TOLD TO DO SO BY THE INVIGILATOR.

NO. of Pages **3**

Question 1 [5 Marks]

Write a method that given a linked list of students, it computes the GPA for each student using only **CSI** courses. Student attributes are:

Name: Student name.

ID: Student ID

Courses_Done: an array of courses done. Stored as course codes e.g CSI247, ISS331 etc.

Courses_Grades: an array of grades corresponding to courses done. A grade is anything between 0 and 5.

Courses_Credits: an array of credits corresponding to courses done.

GPA: student grade point average is computed as $gpa = \frac{\sum_i^n credit \times grade}{n}$. That is sum of credit multiplied by grade divide by number of courses n. For example, if courses A and B have credits 3 and 4 and grades 2.0 and 4.5 respectively, then *the gpa* = *(3x2.0+4x4.5)/2*

Question 2 [6 Marks]

(a) Write a *BubbleSort* method that sorts a given integer array from left to right between indices x and y using recursion. Sort the array in descending order. [4]

(b) What is the main advantage of Selection Sort over Bubble Sort in terms of computational complexity? [2]

Question 3 [7 marks]

a) Write a method that reads a text file line by line and searches for a given string and computes the average number of occurrences of the string per line. [4]

b) Write a method that finds the largest value (assume integer) on the left subtree of a Binary Search Tree implemented using a linked list. Your method should return the value in the node. Assume head pointer is defined.[3]

Question 4 [5 Marks]

Write a method that when given an amount of money in Botswana Pula it gives change in only Botswana coins (P5, P2, P1, P0.50, P0.25, P0.1, P0.05). The method reduces the number of coins in the change by starting with the highest coins first. The method outputs the number of times each coin is used.

Question 5 [12 Marks]

(a) Implement Stack methods: *IsEmpty*, *pop*, *push*, *IsFull* and *peek* using an array. [6]

(b) Write a method that receives a string and randomly exchanges 2 characters of the string. The exchanges are done x number of times, where x is half the length of the string. The method also computes how many characters are different between the original and the new string. [6]

Question 6 [15 Marks]

(a) A directed graph can be represented as an adjacency list in the form of a 2D array. Write a method that converts this adjacency list into an equivalent adjacency matrix. [4]

(b) Given the list of integers below, draw a balanced binary search tree and perform post order traversal on the tree.

45,51,36,17,60,47,49,53,38,47,44,20. [4]

(c) Write a method that takes two 2D arrays A and B. The method multiplies (element by element) elements of row x in A with elements of column y in B to produce a *sum* of those products. The multiplications are performed if the number of columns in A is equal to the number of rows in B. [4]

(d) Convert the expression $A + B * C/F$ to postfix from using a stack. Show all your work. [3]

%%%%%%%% END OF EXAM PAPER %%%%%%%%%%